Hat Guessing with Proper Colorings

Joint Mathematics Meetings (JMM) 2026

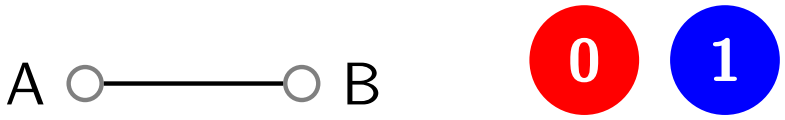
Saptarshi Mandal

Indian Institute of Technology, Kanpur

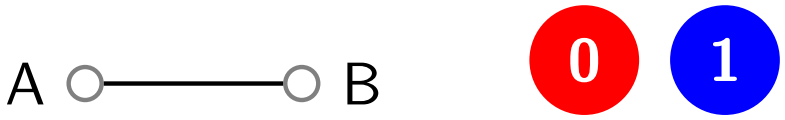
January 4, 2026

Polymath Jr. 2025: Anurag Bishnoi, Anurag Ramachandran, Lars van der Kuil, Michael Kreiger, Peter Bentley, Sam Adriaensen, Saptarshi Mandal

What is the hat guessing game?

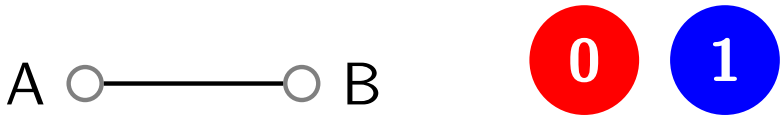


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Alice: I will guess the **same** color that Bob has.
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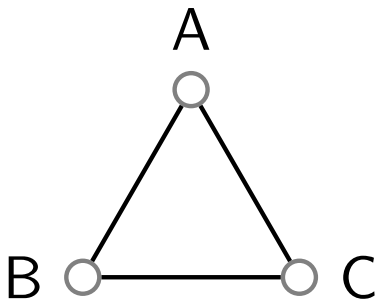
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Alice: I will guess the **same** color that Bob has.
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In other words, if we identify the colors as $\mathbb{Z}_2 = \{0, 1\}$, we have

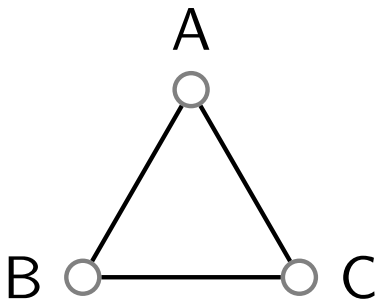
$$g_A = c_B \mod 2$$
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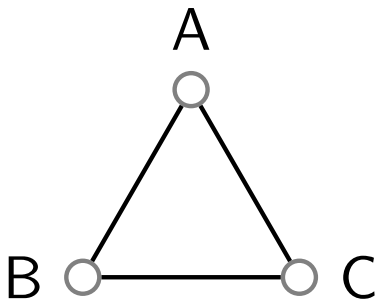




We can identify the colors with the set $\mathbb{Z}_3 = \{0, 1, 2\}$, and the following is a guesing strategy,

$$\begin{aligned} g_A &= 0 - c_B - c_C \mod 3 \\ g_B &= 1 - c_A - c_C \mod 3 \\ g_C &= 2 - c_A - c_B \mod 3 \end{aligned}$$

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In general, if there are n people with n colors and everybody can see eachother's hats, then there is such a strategy.

What is the hat guessing game?

- On a graph $G = (V, E)$, each vertex hosts a player.
- An adversary places on every player, a hat whose color is drawn from a set $[q] := \{0, \ldots, q-1\}$
- Each player sees the hat colors of their neighbors, that is, $N(v)$. They must agree on a strategy: for each $v \in V$, a function

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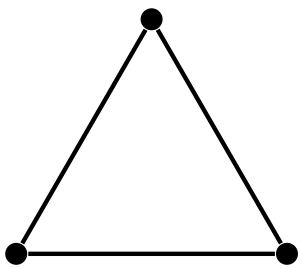
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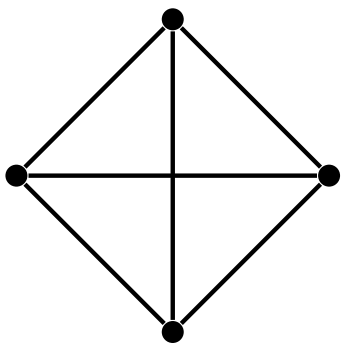
The *proper hat guessing number* of a graph G, $HG_P(G)$, is the largest number of colors q such that a winning strategy exists, GIVEN: the hats always form a proper coloring.

Complete Graphs

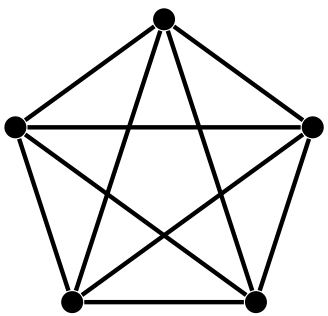
A *complete graph* on n vertices, denoted by K_n is a graph where each vertex has an edge with every other vertex.



K_3



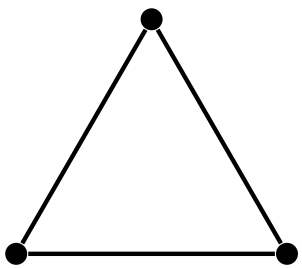
K_4



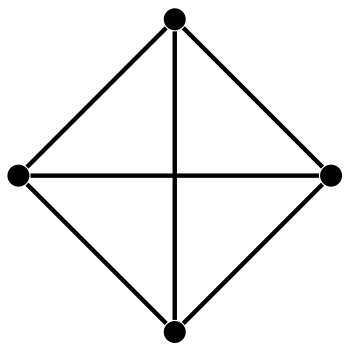
K_5

Complete Graphs

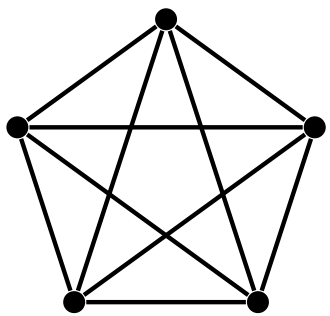
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K_3



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A proper coloring of K_n must assign **distinct** colors to each vertex.

Complete Graphs

Theorem

$HG_P(K_n) = 2n - 1$

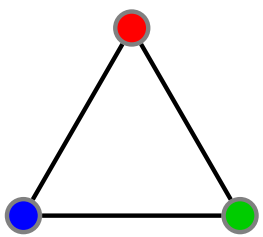
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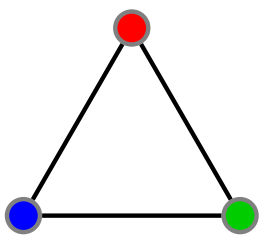


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However, for each vertex $v \in K_3$, exactly 2 distinct colors are visible to it. We can say two things now.

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$$(f_1(c_2, c_3), c_2, c_3) = (c_1, c_2, c_3)$$
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We want to split the set of 60 strings into 3 sets such that for the ith set the ith equality holds. This is possible. In fact, such a splitting automatically defines the f_i's as well.

The Explicit Decomposition ($n = 3$)

$$\text{PROPER} = A_1 \sqcup A_2 \sqcup A_3$$

The set A_1	
012	013
021	041
104	120
124	143
203	231
234	240
301	310
314	342
402	423
430	432

The set A_2	
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In particular, deleting the ith letter of the strings in the ith set, gives all the distinct pairs (the entire domain of f_i).

Complete Graphs

Theorem

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Surprisingly, the proof uses Hall's Marriage Theorem. Define:

$$A := \{\text{strings of } n \text{ distinct letters}\}$$
$$B := \{1, \ldots, n\} \times \{\text{strings of } n-1 \text{ distinct letters}\}$$

where $s \in A$ has an edge with $(i, t) \in B$ if deleting the ith letter of s gives t.

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where $s \in A$ has an edge with $(i, t) \in B$ if deleting the ith letter of s gives t. This n-regular bipartite graph, has a perfect matching, M. We put s in the set A_i where $(s, (i, t)) \in M$. So, $A = \bigsqcup_{i=1}^{n} A_i$.

Complete Graphs

For $n = 5$, we have $2 \cdot 5 - 1 = 9$ colors, namely, $\{0, \ldots, 8\}$.

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The set A for $n = 5$

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The set B for $n = 5$

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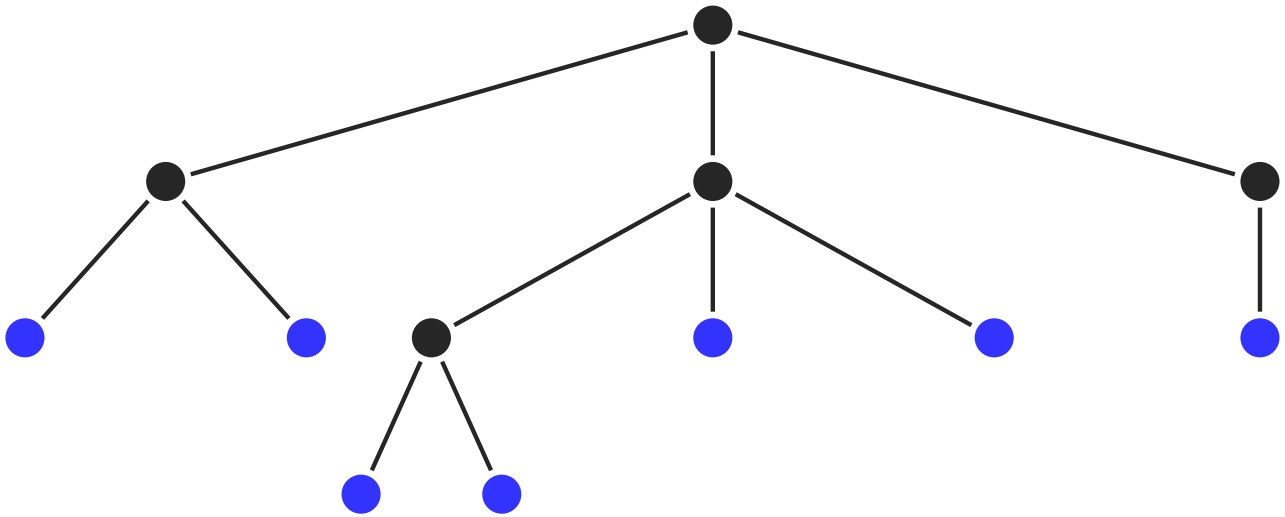
This shows that $HG_P(K_n) \geq 2n - 1$. The upper bound comes from a more general bound that we discuss a few slides later.

Trees

A tree is a connected, acyclic graph. Every tree has atleast 1 leaf.

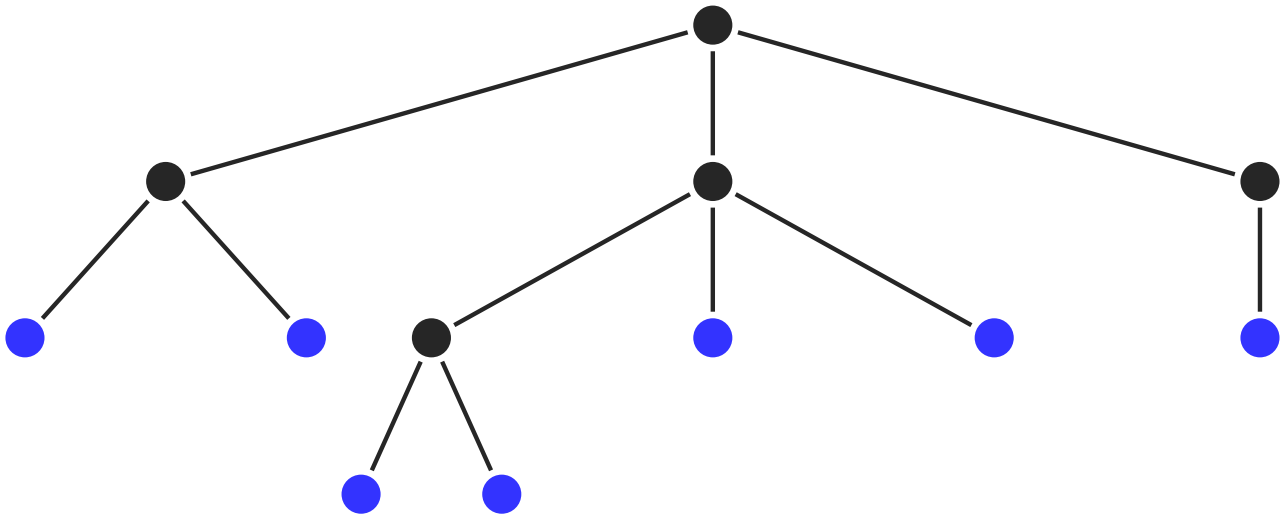
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Theorem

For a connected graph with ≥ 3 vertices, leaf nodes can be removed without reducing HG_P.

Trees

This gives an inductive way to reduce every tree with ≥ 3 vertices to the unique tree on 3 vertices, P_3.



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Base Case

$HG_P(P_3) = 4$

For now, we assume this and state the following.

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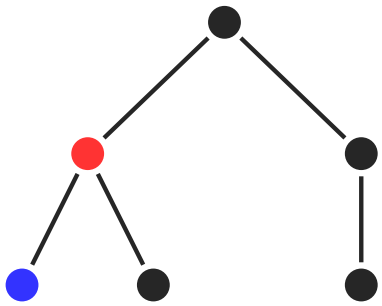
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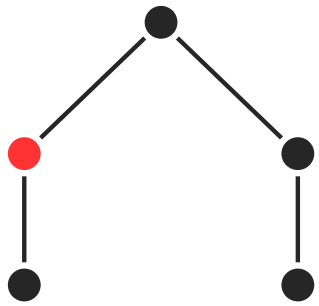
For a tree T on ≥ 3 vertices, $HG_P(T) = 4$.

Trees

Original Tree (T)

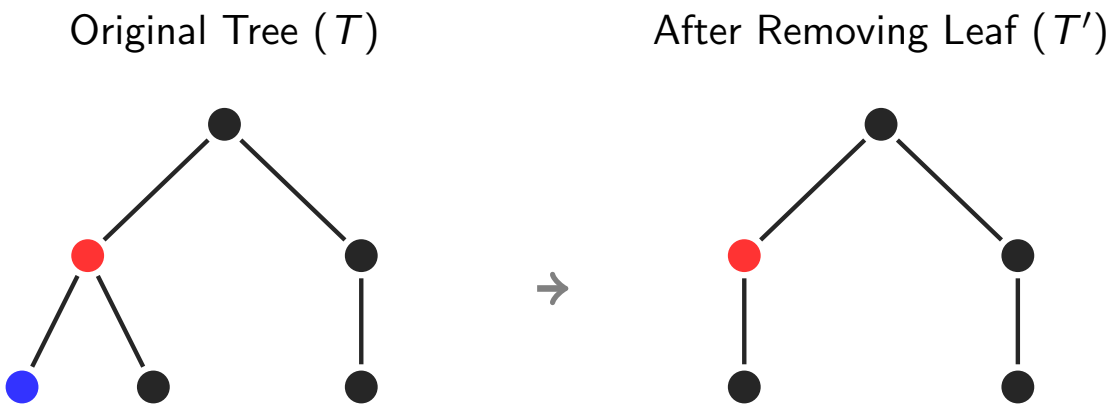


After Removing Leaf (T')



We use a winning strategy f on the original tree T, to create a winning strategy f' for T' on q colors.

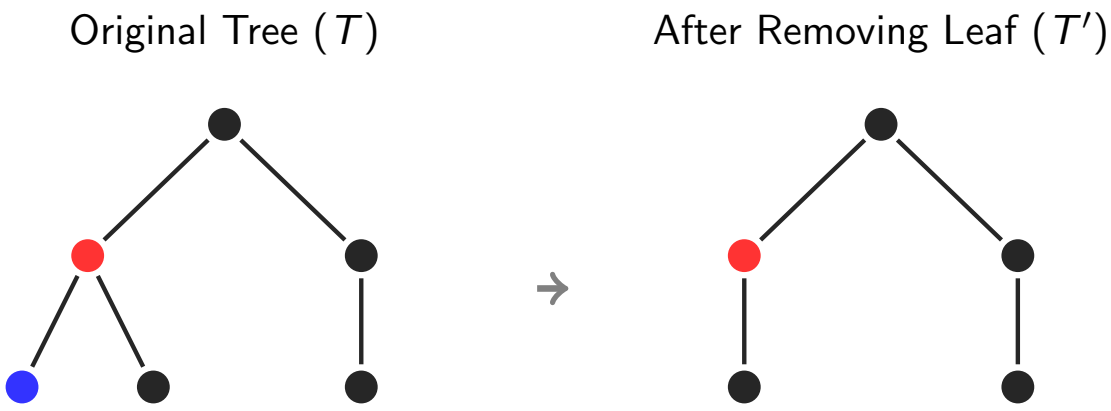
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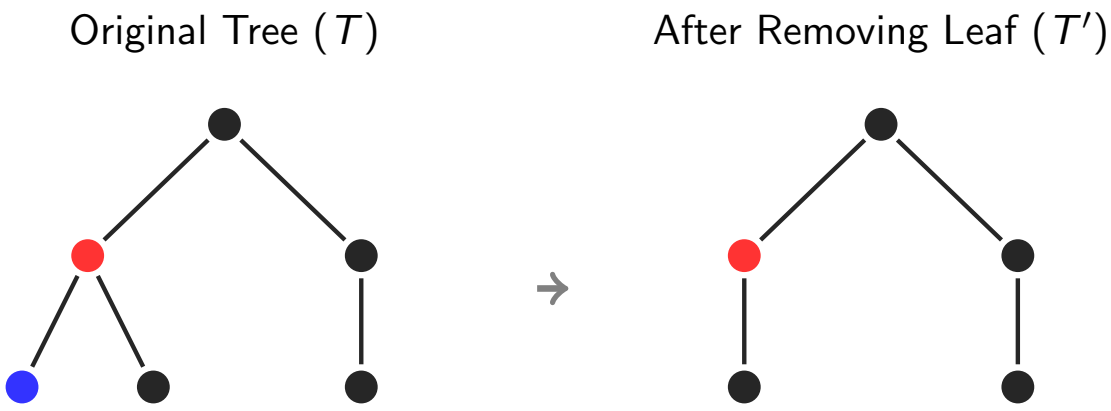
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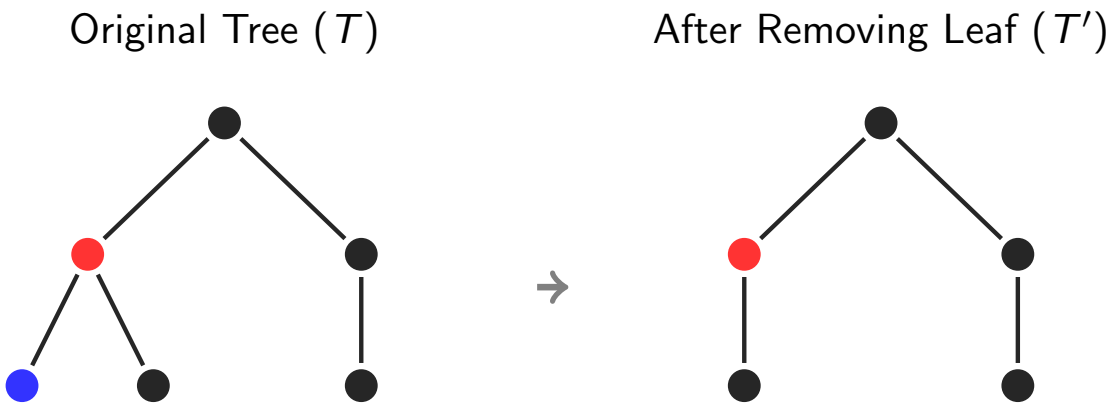


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For all such extensions, either $\bullet$ or $\bullet$ guesses correctly.

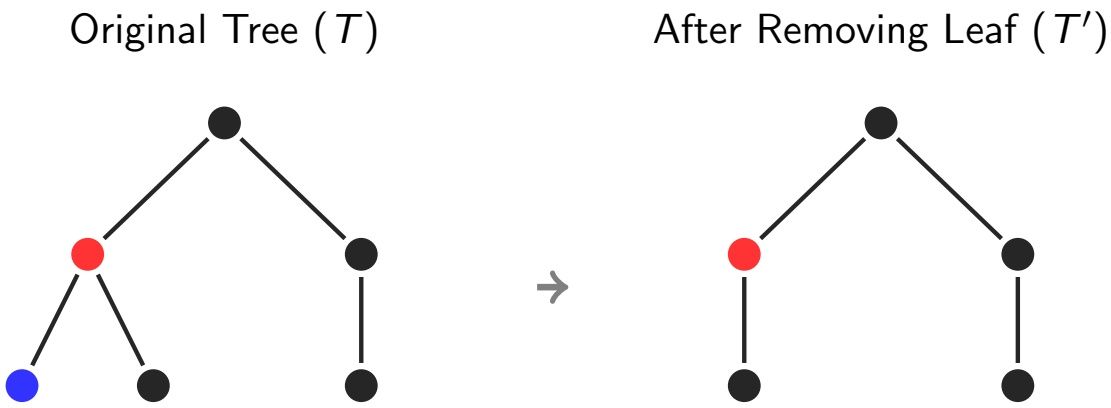
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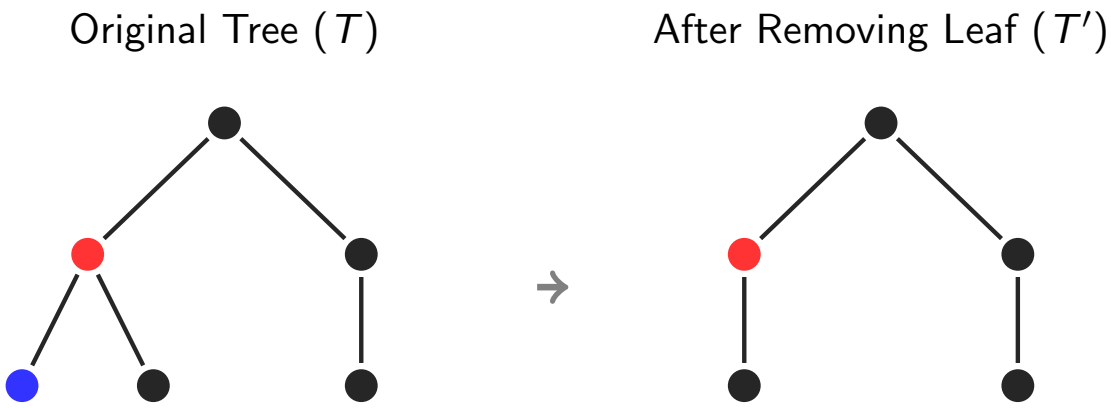
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We define $f'(\bullet) = f(\bullet)$ for the **other** extensions.

General Bounds

Let χ be the chromatic number of a graph G on n vertices.

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$$1 \leq \frac{n}{q-\chi+1} = \sum_{v \in G} \frac{1}{q-\chi+1}$$

as a necessary condition for a winning strategy. So $q \leq n + \chi - 1$.

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$HG_P(G) < \chi(q+1)$ *where* $q = HG(G)$.

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We split the colors into χ families of $q+1$ colors each.

RED 0	BLUE 1	$\cdots$	GREEN $\chi-1$
red 0	blue $(q+1)+0$	$\cdots$	green $(\chi-1)(q+1)+0$
orange 1	cyan $(q+1)+1$	$\cdots$	lime $(\chi-1)(q+1)+1$
brick 2	aqua $(q+1)+2$	$\cdots$	olive $(\chi-1)(q+1)+2$
$\vdots$	$\vdots$	$\ddots$	$\vdots$
maroon q	teal $(q+1)+q$	$\cdots$	emerald $(\chi-1)(q+1)+q$

General Bounds

Theorem

$HG_P(G) < \chi(q+1)$ *where* $q = HG(G)$.

We split the colors into χ families of $q+1$ colors each.

RED 0	BLUE 1	$\cdots$	GREEN $\chi-1$
red 0	blue $(q+1)+0$	$\cdots$	green $(\chi-1)(q+1)+0$
orange 1	cyan $(q+1)+1$	$\cdots$	lime $(\chi-1)(q+1)+1$
brick 2	aqua $(q+1)+2$	$\cdots$	olive $(\chi-1)(q+1)+2$
$\vdots$	$\vdots$	$\ddots$	$\vdots$
maroon q	teal $(q+1)+q$	$\cdots$	emerald $(\chi-1)(q+1)+q$

For a color c (olive), it's family is given by $\lfloor \frac{c}{q+1} \rfloor = \chi - 1$. So, $c = f \cdot (q+1) + j$ where $j \in [q+1]$. We identify $c \sim (f, j)$.

General Bounds

We call a coloring *super*, if no two adjacent vertices have colors from the same family.

$$\text{SUPER} \subseteq \text{PROPER}$$

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For an arbitrary assignment of j-coordinates, there is always a choice of f-coordinates such that the resulting coloring is super. But $j \in [q+1]$ and,

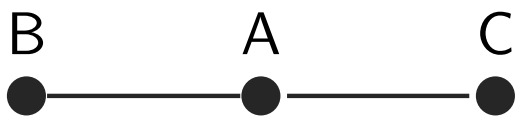
$$HG(G) = q < q + 1$$

Contradiction.

Thank You!

questions?

Appendix A: $HG_P(P_3) = 4$



Guessing function of A

(c_B, c_C)	$f_A(c_B, c_C)$
$(0,0)$ or $(1,0)$	2
$(0,1)$ or $(1,1)$	3
$(0,2)$ or $(1,2)$	3
$(0,3)$ or $(1,3)$	2
$(2,0)$ or $(3,0)$	1
$(2,1)$ or $(3,1)$	0
$(2,2)$ or $(3,2)$	0
$(2,3)$ or $(3,3)$	1

Guessing function of B & C

c_A	$f_B(c_A)$	$f_C(c_A)$
0	1	3
1	0	2
2	3	1
3	2	0

One can verify that this works. :)

Appendix B: Future

Open Q. What is $HG_P(C_n)$?
We know that $HG_P(C_3) = 5$ and $HG_P(C_4) \geq 4$.

Open Q. What is $HG_P(K_{n,n})$?
Inductive approaches? Removing K_2's until we get some base case?